Image Classification

Image Classification

Images

Class Labels

Apple

Pear

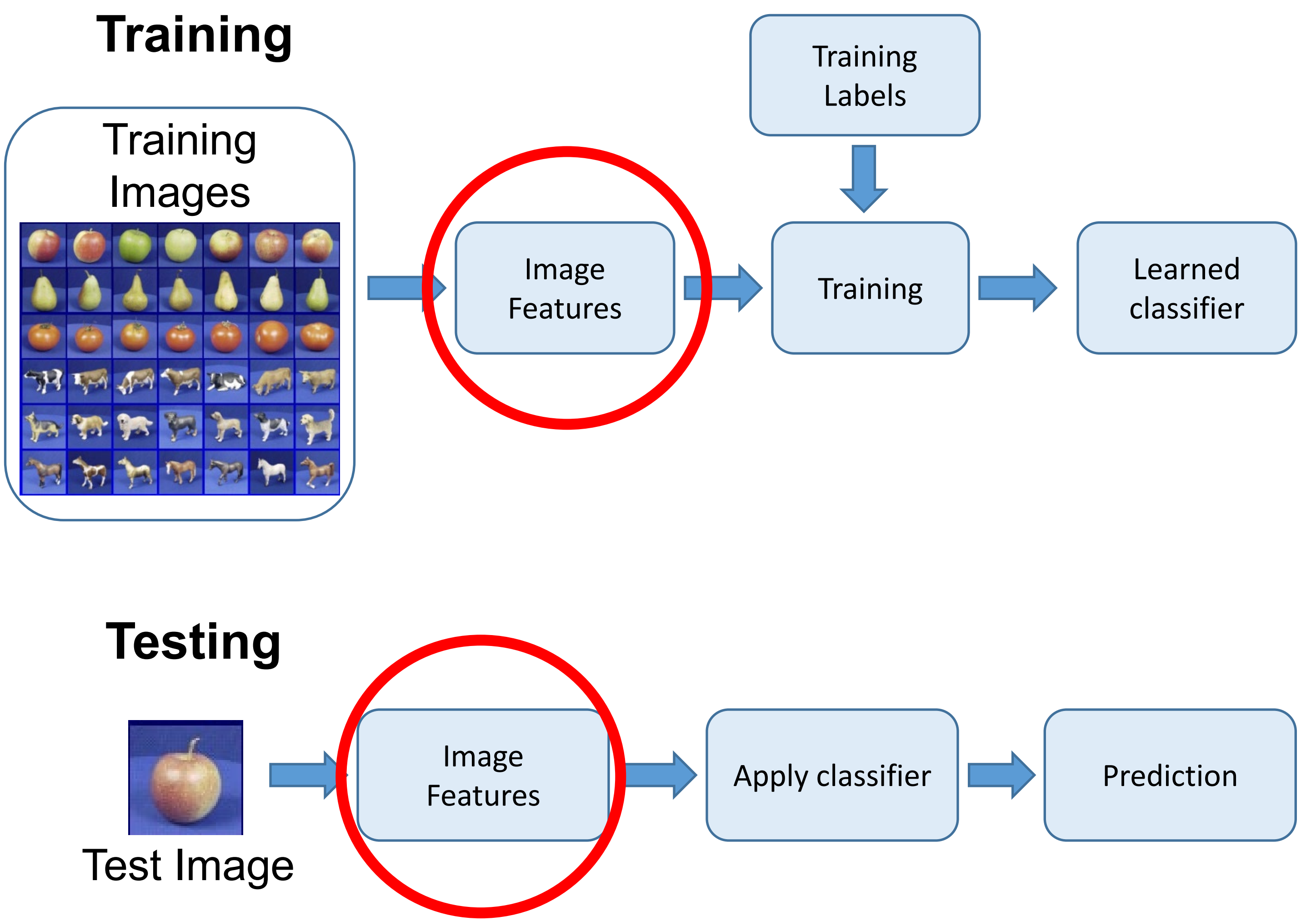
Tomato

Cow

Dog

Horse

Training and Testing Steps



Bags of Words (BoW) for Document Classification

- Frequencies of words from a dictionary are used to represent documents.

Word Clouds

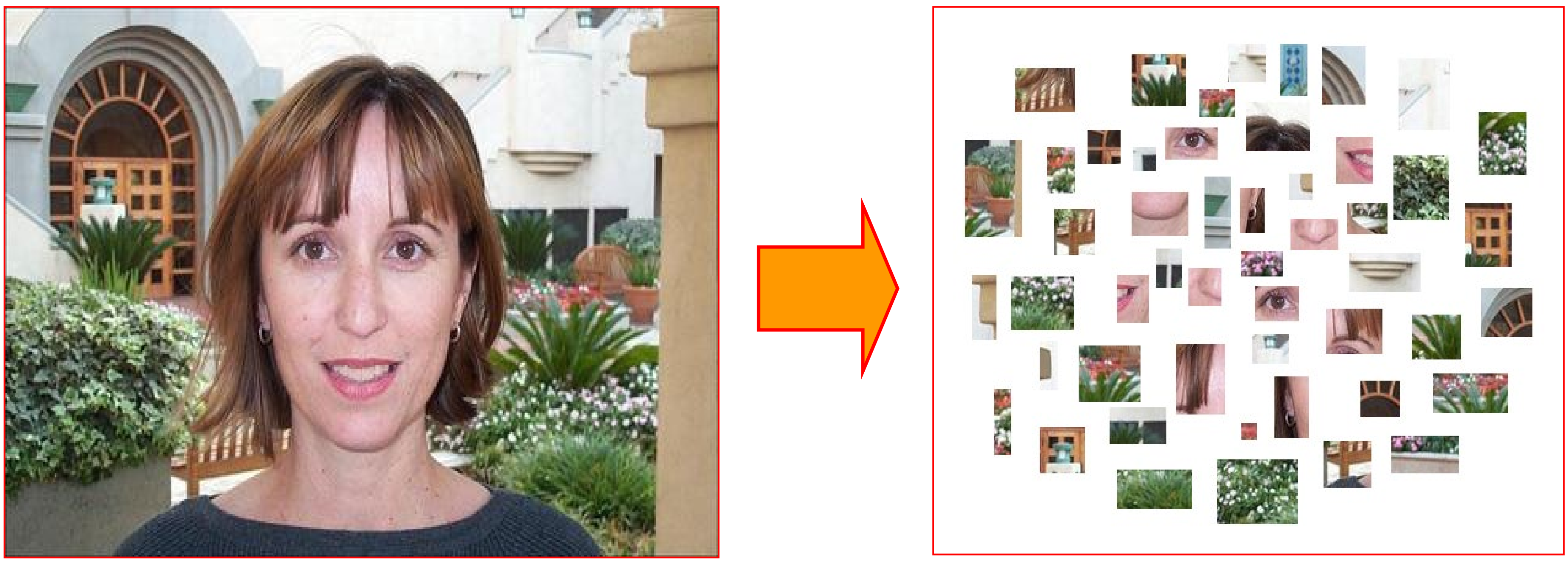
Ref Source: US Presidential Speeches Tag Cloud, https://chir.ag/projects/preztags/

Bags of Words (BoW) for Similar Document Search

- Word frequencies are compared.
- Word order is not considered.

Ref Source: US Presidential Speeches Tag Cloud, https://chir.ag/projects/preztags/

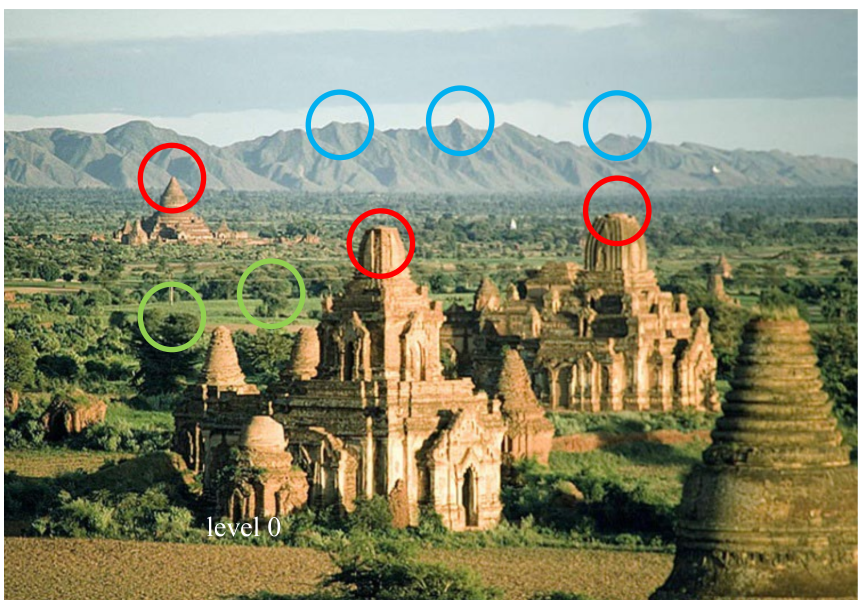
Bag of Visual Words (Features) for Image Classification

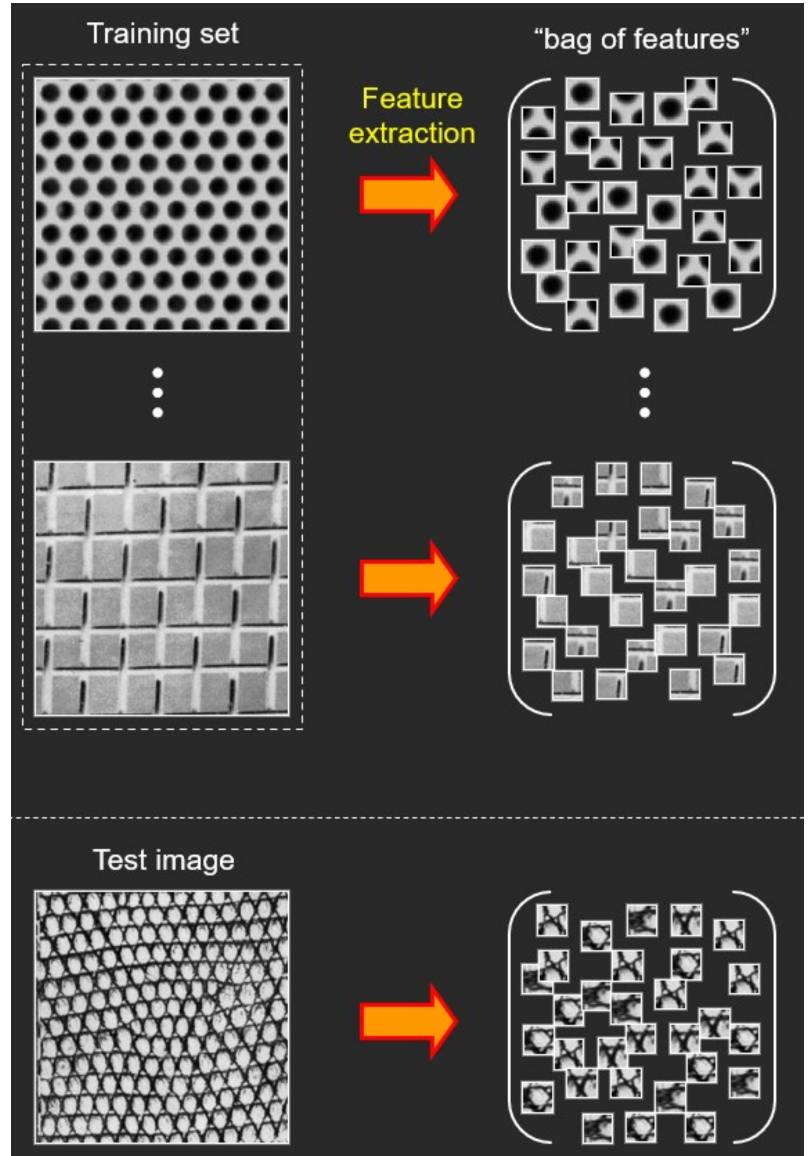


Visual Words or
Image Features

Feature Extraction

- Detecting features (e.g., SIFT) and computing descriptors from images
- Gathering descriptors from images in a collection

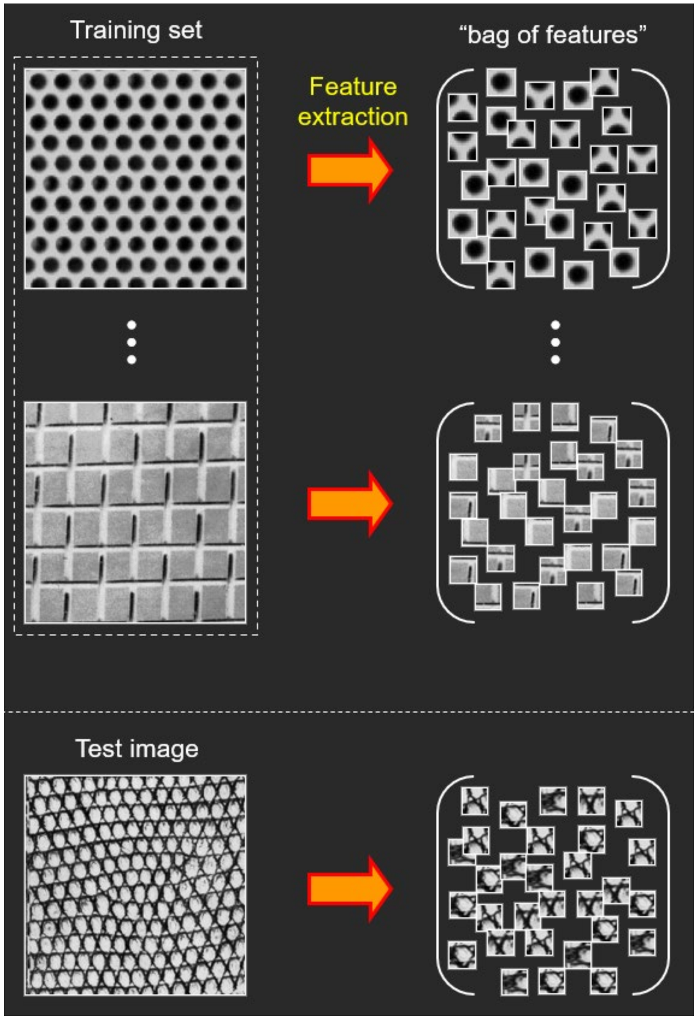


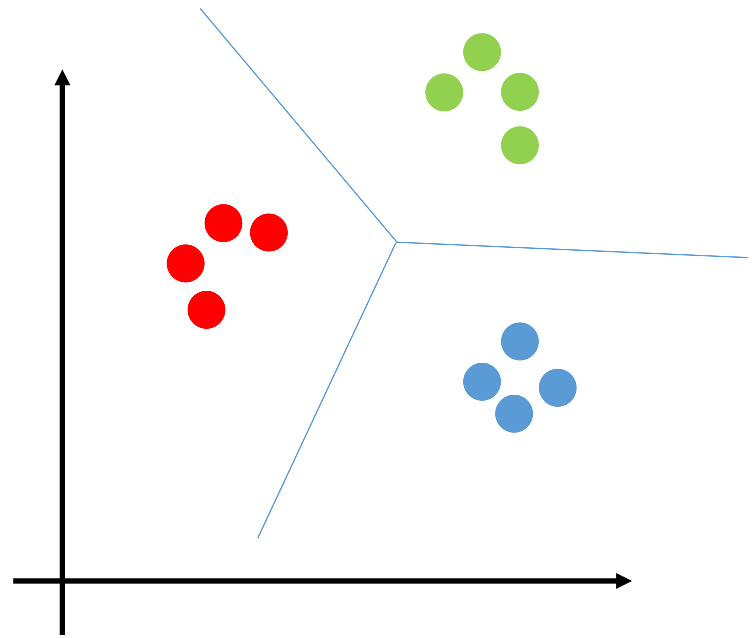


Ref Source: Local, Semi-Local and Global Models for Texture, Object and Scene Recognition, S. Lazebnik. Ph.D. Dissertation, May 2006

Clustering

- Cluster the computed descriptors.





Ref Source: Local, Semi-Local and Global Models for Texture, Object and Scene Recognition, S. Lazebnik. Ph.D. Dissertation, May 2006

K-means Clustering

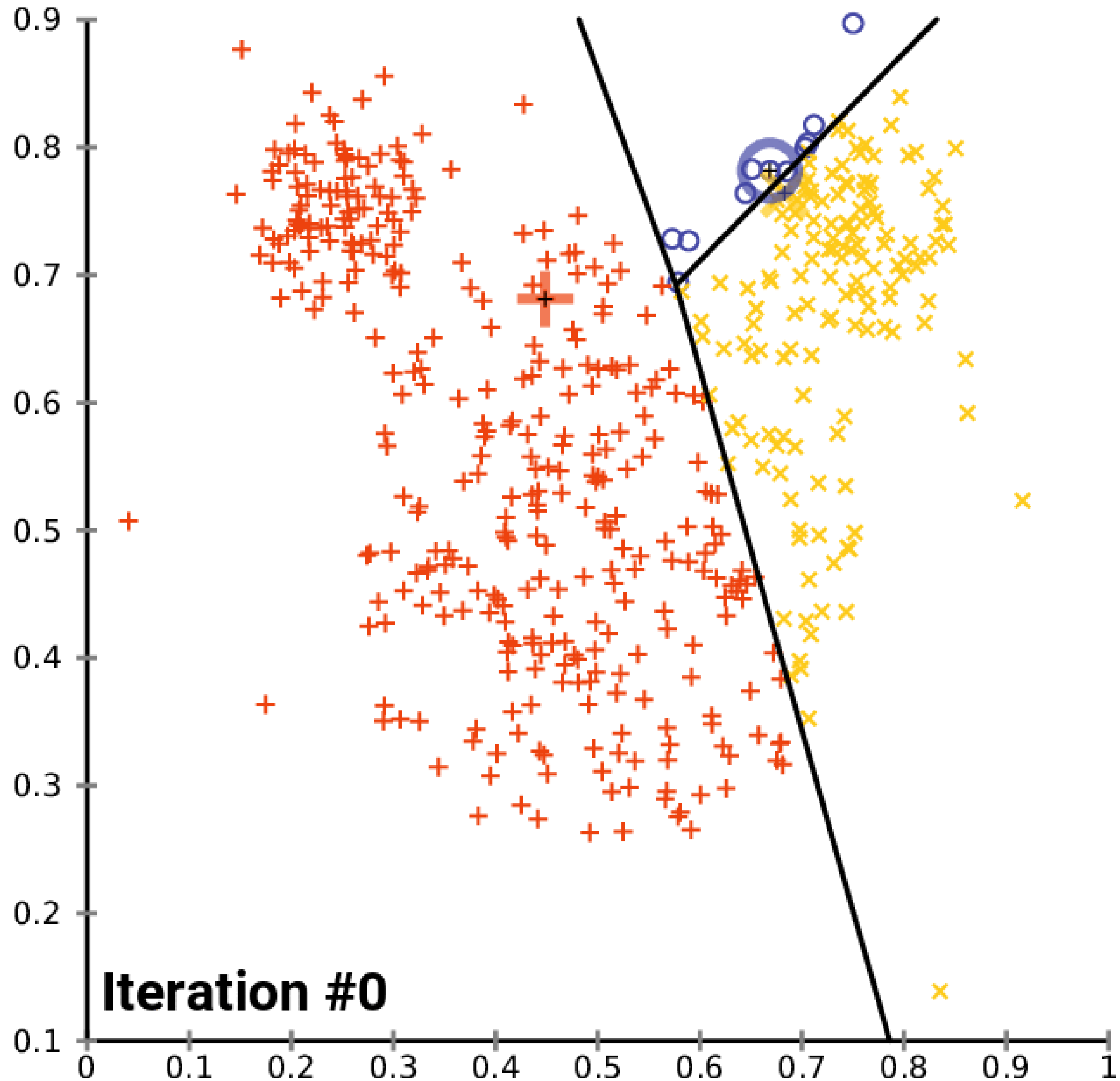
- To partition a set of descriptors/features (samples) into a specified number of disjoint clusters, e.g., k clusters.
- Each descriptor (sample) is assigned to the cluster with the nearest mean, cluster mean.
- The k-means clustering algorithm is an iterative procedure that successfully refines the cluster means until convergence is achieved.

$$\arg\min_{C}\left(\sum_{i=1}^{k}\sum_{z\in C_i}\|\boldsymbol{z}-m_i\|^2\right)$$

where k is the number of clusters, $C=\{C_1,C_2,\dots,C_k\}$ is a list of disjoint clusters, $\boldsymbol{z}$ is the sample vector, m_i is the mean vector.

K-means Clustering

- An example



Ref Source: https://en.wikipedia.org/wiki/K-means_clustering

K-means Clustering

- There are 4 steps.
- [1] Initialize the algorithm: Specify an initial set of means, $m_i(1), i = 1,2,\ldots,k$.
- [2] Assign samples to clusters: Assign each sample to the cluster set whose mean is the closest:

  $\boldsymbol{z}_q \rightarrow C_i \text{ if } \left\|\boldsymbol{z}_q - m_i\right\|^2 < \left\|\boldsymbol{z}_q - m_j\right\|^2, j = 1,2,\ldots,k \ (i \neq j); q = 1,2,\ldots,Q$ (total number of samples)
- [3] Update the cluster centers (means):

$$m_i = \frac{1}{|C_i|} \sum_{\boldsymbol{z} \in C_i} \boldsymbol{z}, i = 1,2,\ldots,k$$

K-means Clustering

- [4] Test for completion: compute the Euclidean norms of the differences between the mean vectors (means) in the current and previous steps. Compute the residual error, E, as the sum of the k norms. Stops if $E \leq T$, where T as specified, nonnegative threshold. Else, go back to Step [2].
- This result at convergence does depend on the initial values chosen for m_i.

K-means Clustering

- For example, if $\boldsymbol{z}$ is an intensity, then it is a segmentation method.

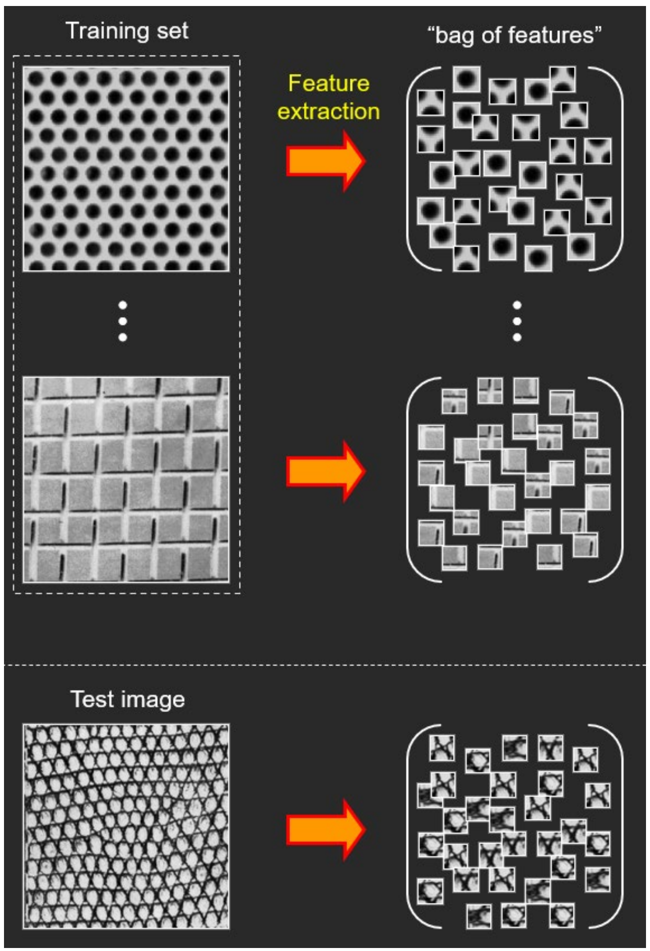
a b

FIGURE 10.49
(a) Image of size 688×688 pixels. (b) Image segmented using the k-means algorithm with $k = 3$.

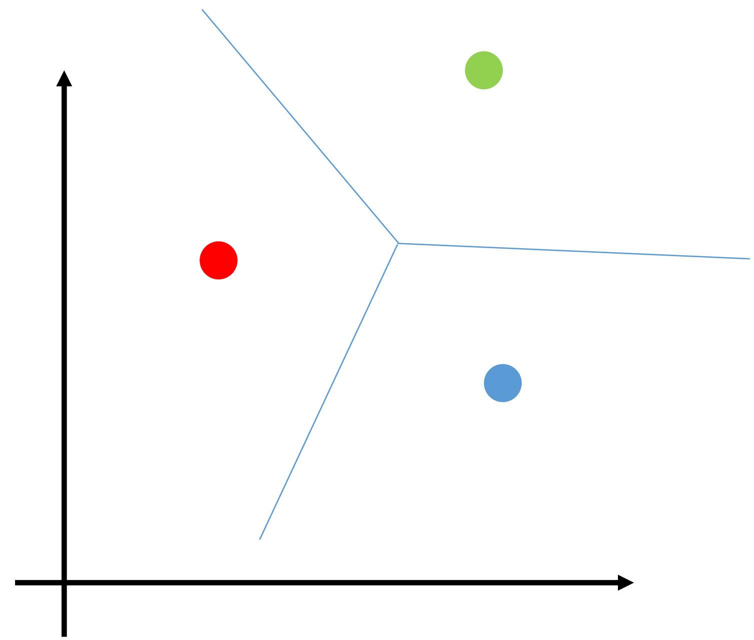
The figure is provided by Pearson Education, Digital Image Processing, Gonzalez & Woods, www.ImageProcessingPlace.com

Visual Vocabulary

- After clustering, the mean vectors are used as the visual words and a visual dictionary is formed (a bag of visual words).



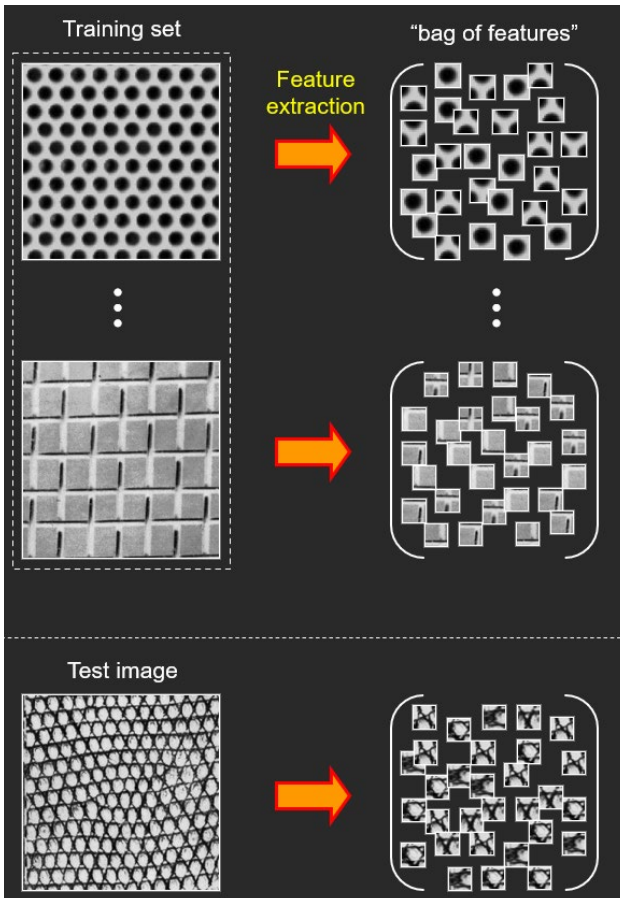
Visual vocabulary

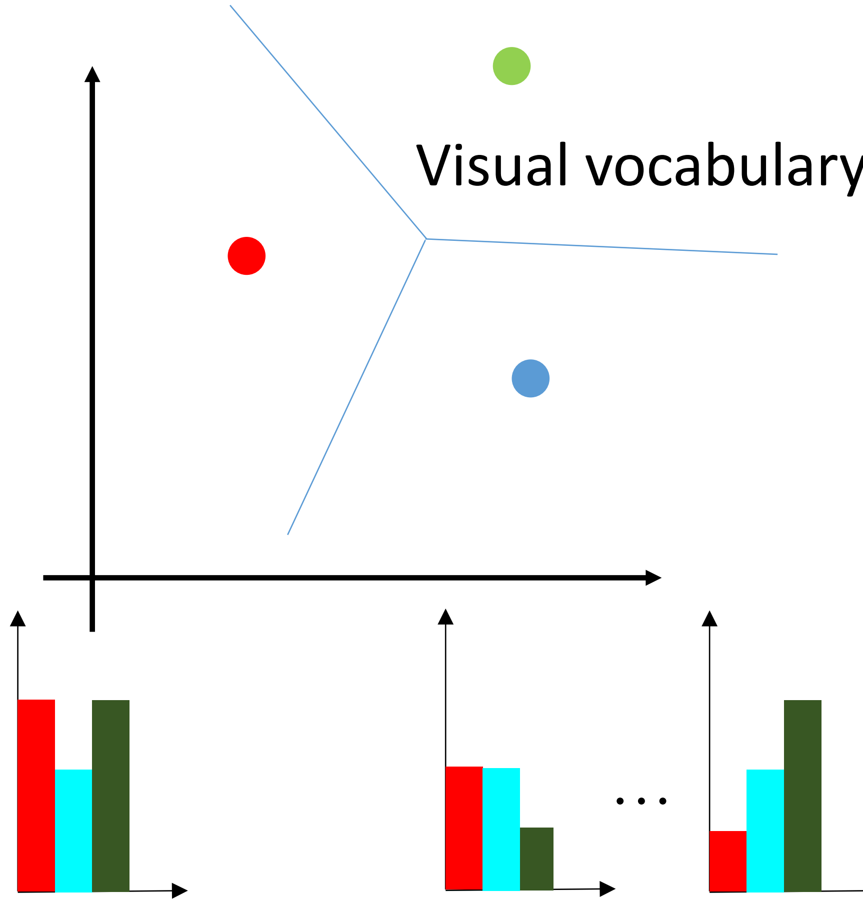


Ref Source: Local, Semi-Local and Global Models for Texture, Object and Scene Recognition, S. Lazebnik. Ph.D. Dissertation, May 2006

A bag of visual words (BoW)

- Pixels are converted into a bag of visual words (a dictionary or a codebook is formed). Images are then represented by frequencies of visual words in the visual vocabulary.





Ref Source: Local, Semi-Local and Global Models for Texture, Object and Scene Recognition, S. Lazebnik. Ph.D. Dissertation, May 2006

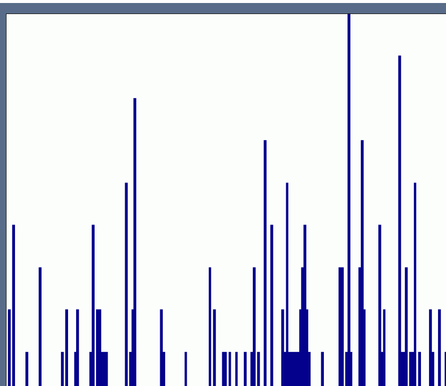
Spatial Pyramid

- The content can be inferred from the frequencies of visual words that happen in an image, i.e., a histogram of visual words is formed.

- Spatial pyramid can be used to capture the spatial relationship between visual words.

- A new histogram can be formed by concatenating the individual histograms. The newly formed histogram can then be used to present the image for classification.

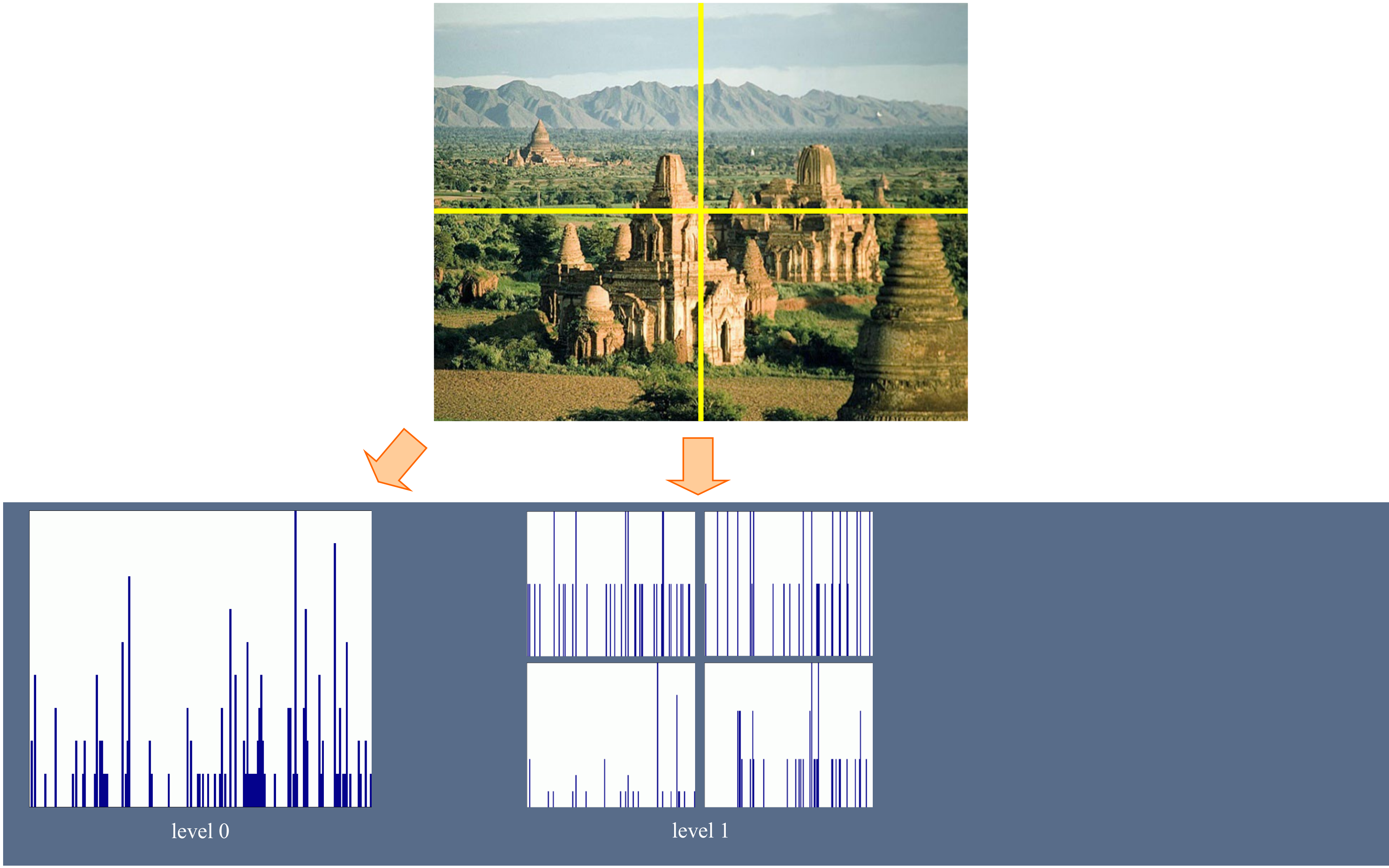
Spatial Pyramid



level 0

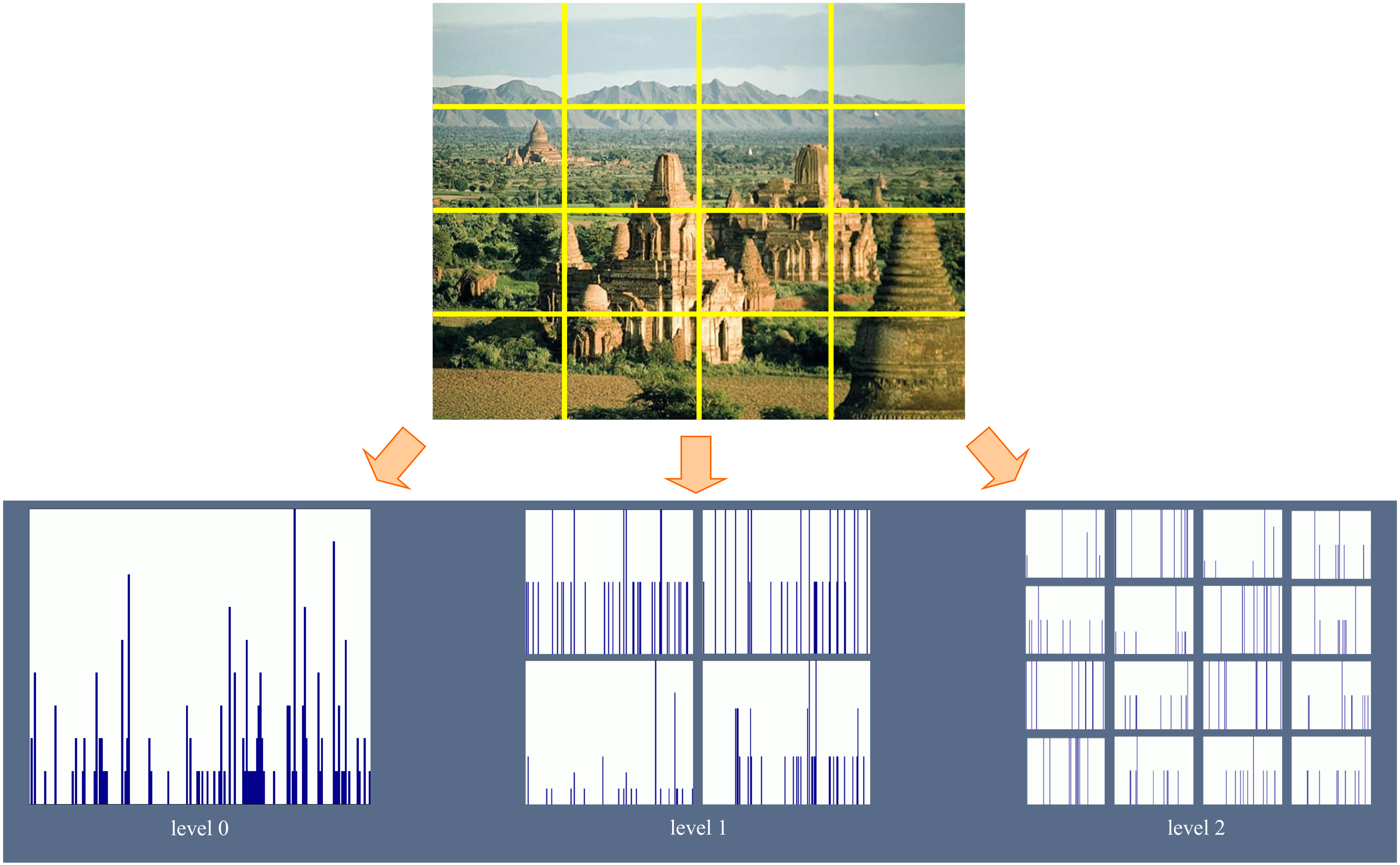
Lazebnik et al., Beyond Bags of Features: Spatial Pyramid Matching for Recognizing Natural Scene Categories, CVPR, 2006.

Spatial Pyramid



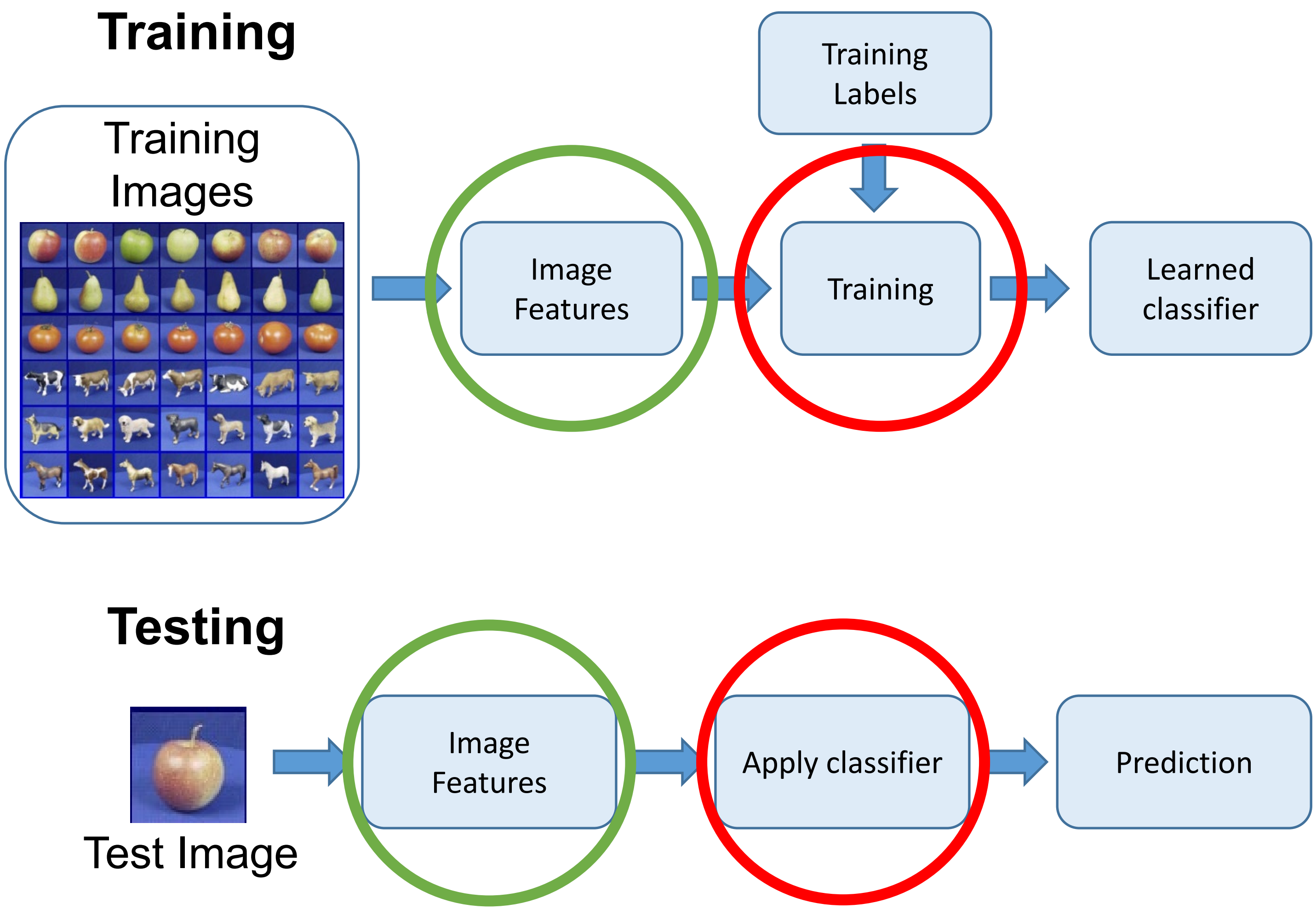
Lazebnik et al., Beyond Bags of Features: Spatial Pyramid Matching for Recognizing Natural Scene Categories, CVPR, 2006.

Spatial Pyramid



Lazebnik et al., Beyond Bags of Features: Spatial Pyramid Matching for Recognizing Natural Scene Categories, CVPR, 2006.

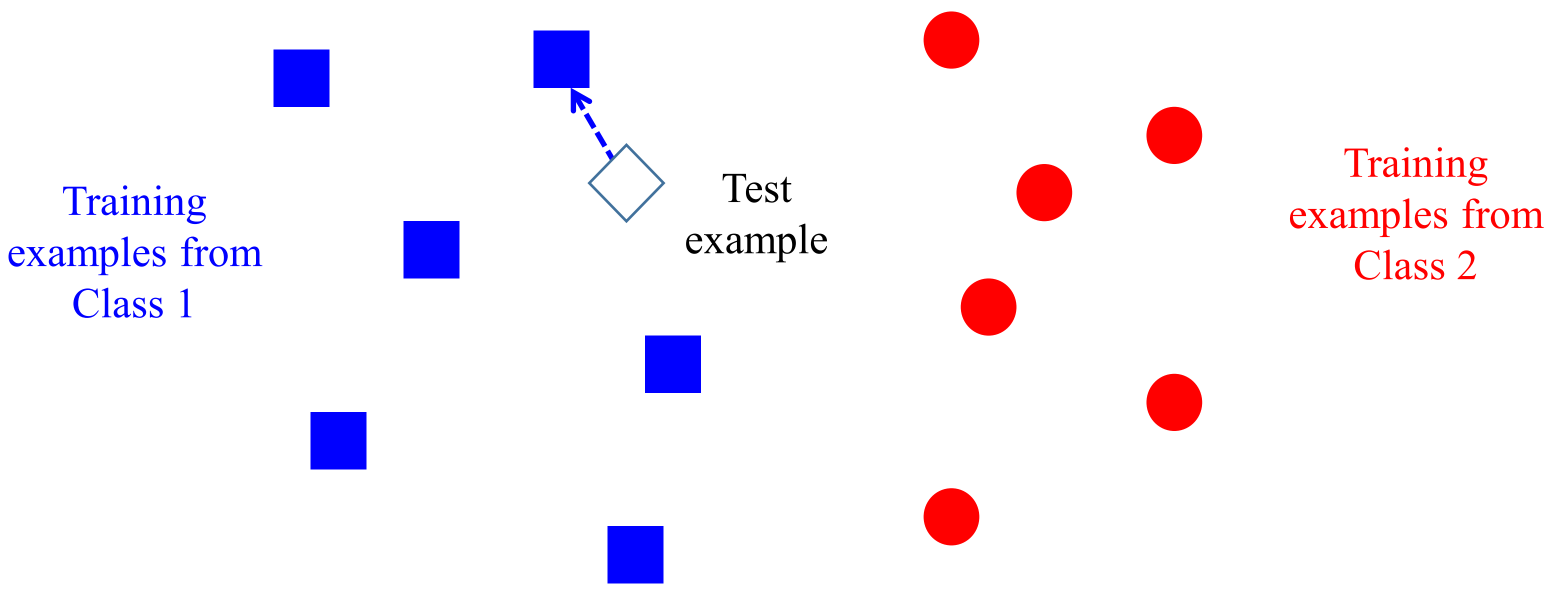
Training and Testing Steps



Classifier

Nearest Neighbor Classifier

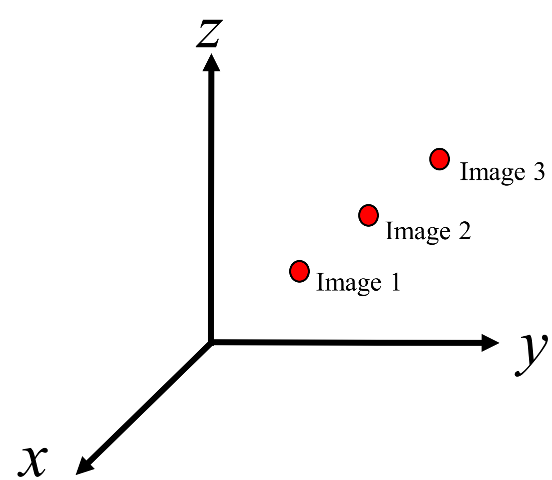
- f(**x**) = label of the training example nearest to **x**, which is the image presentation.
- All we need is a distance or similarity function for the inputs.
- The label is assigned by searching the nearest neighbor among the examples.
- Other classifiers can also be considered, e.g., FLD, SVM.



Dimensionality Reduction for Image Classification

- The number of dimensions in the image representation can be reduced by using PCA (Principal Component Analysis). PCA is a method for dimensionality reduction of the data, i.e., image representation.
- One application is face recognition (identification).
- For example, if there are three 3x1 images (see below), then each image has three intensity values. If each intensity value is treated as a coordinate in 3D, then each image can be represented as a point in 3D.

	Image 1	Image 2	Image 3
x	1	2	3
y	2	4	6
z	3	6	9

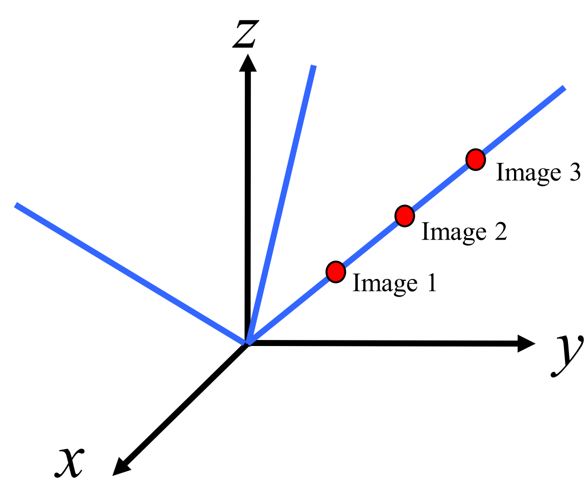


Stan Z. Li and Anil K. Jain, Handbook of Face Recognition, Springer, 2005, and notes by Robert Collins, PSU

Dimensionality Reduction for Image Classification

- Consider a new coordinate system in which one of the axes passes through these points. Then each point can be represented by a single value. The coordinate values in the other axes become zero or very small.
- To represent these points effectively, the number of dimensions can be reduced from three to one.

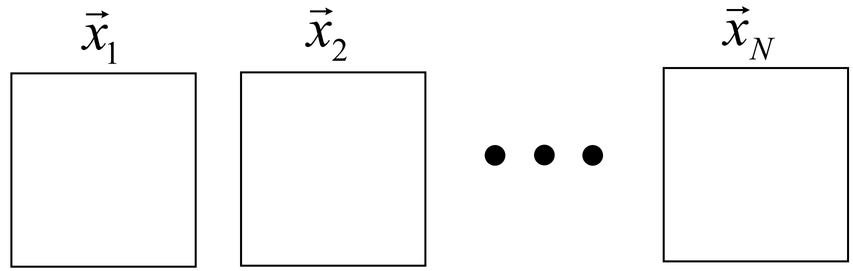
	Image 1	Image 2	Image 3
x	1	2	3
y	2	4	6
z	3	6	9



Stan Z. Li and Anil K. Jain, Handbook of Face Recognition, Springer, 2005, and notes by Robert Collins, PSU

Eigenfaces

- Let us consider a set of *N* sample images (image vectors) with *m* x *n* dimensions



- Each image is represented by a 1D vector with dimensions (*m* x *n*) x 1.

$$\vec{x}_1 = \begin{bmatrix} x_{1,1} \\ x_{1,2} \\ \vdots \\ x_{1,mn} \end{bmatrix}$$

- The mean image vector is given by

$$\bar{x} = \frac{1}{N} \sum_{i=1}^{N} \begin{bmatrix} x_{i,1} \\ x_{i,2} \\ \vdots \\ x_{i,mn} \end{bmatrix}$$

Peter N. Belhumeur, Joao P. Hespanha, David J. Kriegman, Eigenfaces vs. Fisherfaces: Recognition using class specific linear project, IEEE Transactions on Pattern Analysis and Machine Intelligence, 19(7), 711-720, 1997, and notes by Robert Collins, PSU

Eigenfaces

- The scatter matrix is given by

$$\overleftrightarrow{S} = \left[\vec{x}_1 - \overline{x} \quad \vec{x}_2 - \overline{x} \quad \cdots \quad \vec{x}_N - \overline{x} \right] \begin{bmatrix} \left(\vec{x}_1 - \overline{x} \right)^T \\ \left(\vec{x}_2 - \overline{x} \right)^T \\ \vdots \\ \left(\vec{x}_N - \overline{x} \right)^T \end{bmatrix}$$

- The corresponding t eigenvectors with non-zero eigenvalues λ_i are

$$\vec{e}_1 \quad \vec{e}_2 \quad \cdots \quad \vec{e}_t$$

where $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_t$

- Then,

$$\vec{x}_j \approx \overline{x} + \sum_{i=1}^{t} g_{ji} \vec{e}_i \qquad \text{where} \qquad g_{ji} = \left(\vec{x}_j - \overline{x} \right) \cdot \vec{e}_i$$

Peter N. Belhumeur, Joao P. Hespanha, David J. Kriegman, Eigenfaces vs. Fisherfaces: Recognition using class specific linear project, IEEE Transactions on Pattern Analysis and Machine Intelligence, 19(7), 711-720, 1997, and notes by Robert Collins, PSU

Eigenfaces

- Since the eigenvectors have the same dimension as the image vectors, the eigenvectors are referred to as Eigenfaces.
- The value of t is usually much smaller than the value of $m \times n$. Therefore, the number of dimensions can be reduced significantly.
- For example, there are 20 face images. The mean image and its first three eigenfaces are shown below.

$\bar{x}$ Mean image | $\vec{e}_1$ 1st Eigenface | $\vec{e}_2$ 2nd Eigenface | $\vec{e}_3$ 3rd Eigenface

…

Peter N. Belhumeur, Joao P. Hespanha, David J. Kriegman, Eigenfaces vs. Fisherfaces: Recognition using class specific linear project, IEEE Transactions on Pattern Analysis and Machine Intelligence, 19(7), 711-720, 1997, and notes by Robert Collins, PSU
Some of the images can be found at http://cswww.essex.ac.uk/mv/allfaces/index.html , http://peipa.essex.ac.uk/benchmark/databases/

Eigenfaces

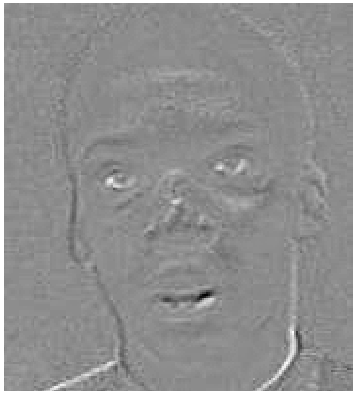
- Given the average image and the eigenfaces, an image can be represented by a linear combination of the average image vector and eigenfaces.

$$\vec{x} \approx \overline{x} + \sum_{i=1}^{t} g_i \vec{e}_i$$

Mean image $\overline{x}$

$\vec{x} \approx$ 

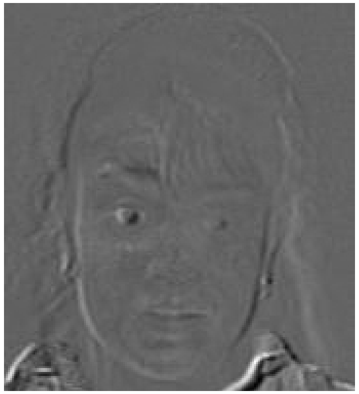
1st Eigenface $\vec{e}_1$

$+g_1$ 

2nd Eigenface $\vec{e}_2$

$+g_2$ 

3rd Eigenface $\vec{e}_3$

$+g_3$ 

$+\cdots+ g_t\vec{e}_t$

- The values of *g*'s become the new representation of the above image for classification (face recognition).

Peter N. Belhumeur, Joao P. Hespanha, David J. Kriegman, Eigenfaces vs. Fisherfaces: Recognition using class specific linear project, IEEE Transactions on Pattern Analysis and Machine Intelligence, 19(7), 711-720, 1997, and notes by Robert Collins, PSU
Some of the images can be found at http://cswww.essex.ac.uk/mv/allfaces/index.html , http://peipa.essex.ac.uk/benchmark/databases/

Eigenfaces

- Process the image database
  - Pre-process the images, e.g., histogram equalization
  - Run PCA and compute eigenfaces
  - Calculate *t* coefficients for each image, $g_1, g_2, \ldots, g_t$
- Given a new image to be recognized $\vec{x}$
  - Calculate *t* coefficients for the new image

$$\left(g_1, g_2, \cdots, g_t\right)$$

- Detect if the new image is a face

$$\left\| \vec{x} - \left(\overline{x} + g_1 \vec{e}_1 + g_2 \vec{e}_2 \cdots + g_t \vec{e}_t \right) \right\| < \text{Threshold}$$

- If it is a face, find the closest labeled face based on the nearest neighbor in the *t*-dimensional space.

Peter N. Belhumeur, Joao P. Hespanha, David J. Kriegman, Eigenfaces vs. Fisherfaces: Recognition using class specific linear project, IEEE Transactions on Pattern Analysis and Machine Intelligence, 19(7), 711-720, 1997, and notes by Robert Collins, PSU